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Effects of rumen-protected arginine (RPA) on uterine metabolome in dairy cows

Objectives were to determine the effects of supplying 30 g/d of RPA from 250 d of gestation to 21 d in milk (**DIM**) on the uterine metabolome on day 16 of pregnancy. Holstein cows (n=102) were assigned in a randomized complete block design to control (**CON**; 30 g/d of metabolizable protein from heat-treated soybean meal) or RPA (30 g/d of metabolizable arginine; Ajinomoto Co., Inc., Japan). Estrous cycle was synchronized and cows received time artificial insemination (**AI**) at 58 DIM. The uterine horn ipsilateral to the corpus luteum was sampled for metabolome analyses using a cytobrush and subsequently flushed for conceptus collection on d 16 after AI. Cows with a recovered conceptus or with interferon tau (**IFNT**) >200 pg/mL were considered pregnant. The uterine metabolome from 38 pregnant cows (19 CON and 19 RPA) was analyzed using the MxP Quant500 kit. Data were analyzed by linear mixed-effects models using MIXED or GLIMMIX procedures in SAS/STAT, whereas metabolomic data were analyzed in MetaboAnalyst 6.0 with adjustments for parity and sampling cohort. Supplementing RPA did not affect pregnancy on day 16 after AI, conceptus length, and IFNT content in the uterine fluid. Metabolomic analysis revealed 73 compounds that tended to differ between treatments (54 increased and 19 decreased) after adjustment for false discovery. Supplementing RPA increased the concentrations of some triacylglycerols and cholesteryl esters, asparagine, and spermidine. Pathway enrichment analysis indicated RPA influenced ($P<0.03$) primarily amino acids metabolism (aspartate and methionine), bile acid biosynthesis, ammonia recycling, and spermidine and spermine biosynthesis. Because RPA supplementation stopped at 21 DIM, it is likely that the effects of treatment on the uterine metabolome were consequences of changes that took place in early lactation.

Item	CON	RPA	SEM	<i>P</i> -value
Pregnant d 16, %	66.7	72.8	8.3	0.59
Conceptus length, cm	11.5	10.8	1.3	0.70
IFNT, ng/mL	2.18	3.09	0.75	0.38
Metabolites, μ M				
Triacylglycerols	8.0	23.4	6.0	0.06
Cholesterol esters	2.1	3.2	0.4	0.03
Spermidine	0.05	0.26	0.07	0.02
Asparagine	4.4	8.4	1.2	0.02

Arginine, conceptus, uterine metabolome

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